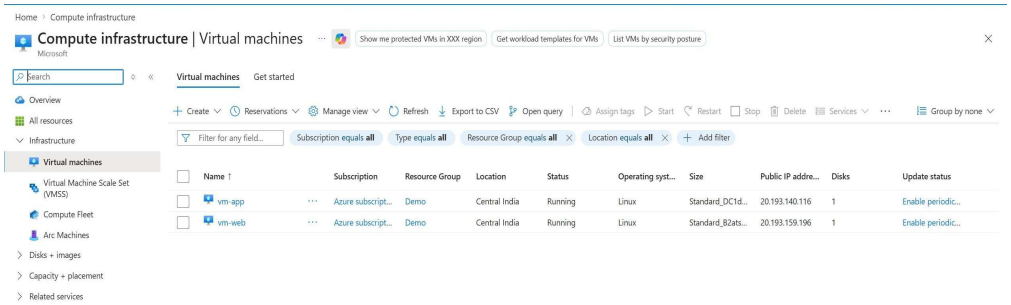
Day 78

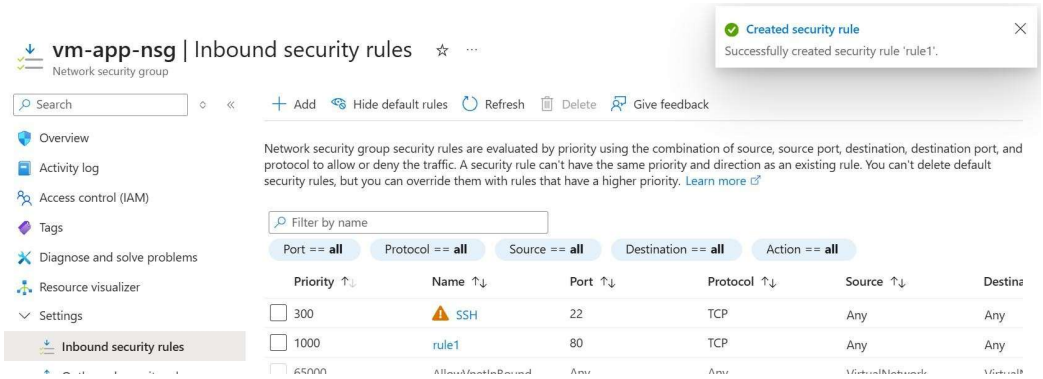
Step 1: Create Virtual Machines



Step 2: Verify NGINX is running on VM

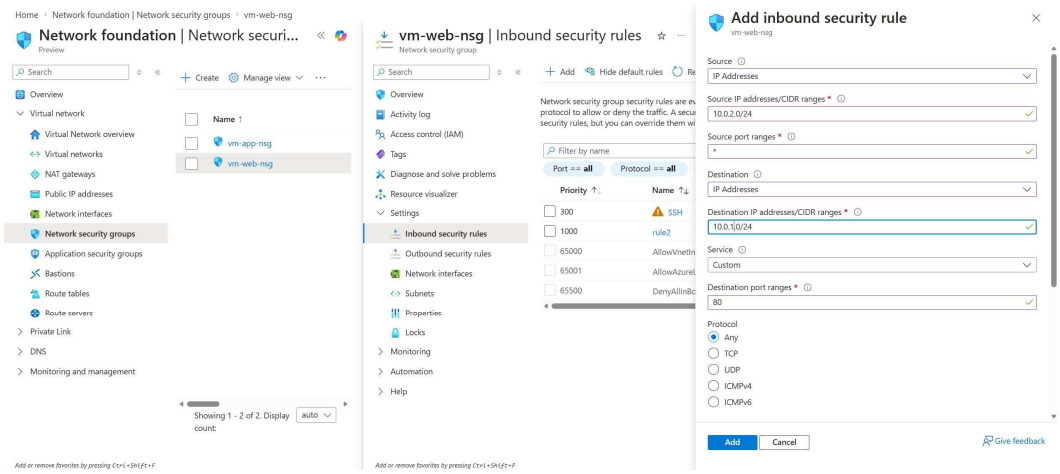
```
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@vm-app:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-04-10 15:15:31 UTC; 2min 14s ago
       Docs: man:nginx(8)
    Process: 2398 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 2400 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 2428 (nginx)
      Tasks: 2 (limit: 9504)
     Memory: 1.7M (peak: 3.8M)
        CPU: 11ms
     CGroup: /system.slice/nginx.service
             ├─2428 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─2430 "nginx: worker process"

Apr 10 15:15:31 vm-app systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Apr 10 15:15:31 vm-app systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
azureuser@vm-app:~$
```

Step 3: NSG (Network Security Group)

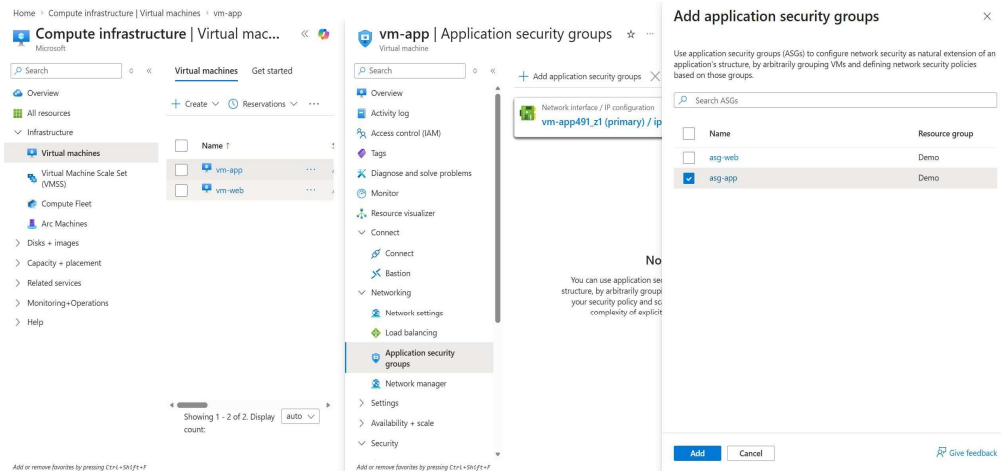


Step 4: nginx should load

Step 5: Secure Network Traffic (Real Scenario)



Step 6: Application Security Groups (ASG)



```
azureuser@vm-app:~$ ping 10.0.0.4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=0.640 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=0.624 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=0.598 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=0.590 ms
64 bytes from 10.0.0.4: icmp_seq=5 ttl=64 time=0.674 ms
64 bytes from 10.0.0.4: icmp_seq=6 ttl=64 time=1.09 ms
64 bytes from 10.0.0.4: icmp_seq=7 ttl=64 time=0.561 ms
64 bytes from 10.0.0.4: icmp_seq=8 ttl=64 time=0.477 ms
64 bytes from 10.0.0.4: icmp_seq=9 ttl=64 time=0.471 ms
64 bytes from 10.0.0.4: icmp_seq=10 ttl=64 time=0.489 ms
64 bytes from 10.0.0.4: icmp_seq=11 ttl=64 time=0.579 ms
64 bytes from 10.0.0.4: icmp_seq=12 ttl=64 time=4.50 ms
64 bytes from 10.0.0.4: icmp_seq=13 ttl=64 time=3.47 ms
64 bytes from 10.0.0.4: icmp_seq=14 ttl=64 time=1.33 ms
64 bytes from 10.0.0.4: icmp_seq=15 ttl=64 time=0.669 ms
64 bytes from 10.0.0.4: icmp_seq=16 ttl=64 time=0.562 ms
^C
--- 10.0.0.4 ping statistics ---
16 packets transmitted, 16 received, 0% packet loss, time 15317ms
rtt min/avg/max/mdev = 0.471/1.082/4.500/1.132 ms
azureuser@vm-app:~$
```